

Pandrosion Relativity Benchmarks

Minkowski recovery, Newtonian shadow, and horizon-safe dynamic atlases

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Abstract

Paper 027 formulated Pandrosion relativity as a finite-probe version of relativistic geodesic motion: causal candidate endpoints, a discrete proper-time score, and dynamic atlas selection. The present paper turns that formulation into reproducible numerical tests. Three benchmarks are built. First, in $1 + 1$ Minkowski space, a causal graph maximizer recovers the straight inertial worldline and its proper time. Second, the weak-field order- c^{-2} Pandrosion score recovers Newton's equation; for a constant field the selected bridge converges to the classical parabola. Third, for radial infall near a Schwarzschild horizon, the dynamic atlas rejects the ill-conditioned Schwarzschild chart and switches to a regular Painleve–Gullstrand chart before the horizon.

The experiments are intentionally conservative. They do not claim a new relativistic law. The novelty is the Pandrosion organization of the calculation: finite causal probes, proper-time maximization, weak-field Newtonian shadow, and chart-conditioning logic are tested as one algorithmic pipeline. In the finest runs reported here, the Minkowski path error is 0.00500 at mesh $\Delta x = 0.01250$, the Newtonian bridge has RMS error 0.00851 at mesh $\Delta x = 0.00500$, and the horizon atlas switches to the regular chart at $r \simeq 2.603M$.

One-sentence summary. 028 makes 027 operational: finite Pandrosion probes reproduce the inertial line, the Newtonian parabola, and a horizon-safe chart switch in explicit computations.

Is it new? The physical laws and many numerical ingredients are classical. What is new inside this Pandrosion program is the combined protocol: causal finite-slope probe graphs, proper-time scoring, a Newtonian shadow test, and dynamic atlas conditioning presented as one Pandrosion benchmark suite.

Reproducibility. All figures and numerical constants in this paper are generated by

`scripts/028_generate_pandrosion_relativity_benchmarks.py`.

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1 Why a benchmark paper is needed

Paper 027 gave the geometric formulation. A formulation is not enough. A Pandrosion physical law should pass minimal tests:

- (i) in flat spacetime it must recover inertial motion;
- (ii) in the weak-field slow-motion limit it must recover Newton's equation;
- (iii) near a coordinate singularity it must switch charts instead of treating the coordinate blow-up as physical.

The three tests below are designed exactly for those checks. They are small, controlled, and reproducible.

2 Benchmark 1: Minkowski causal graph

Use $1 + 1$ Minkowski spacetime with units $c = 1$:

$$ds^2 = -dt^2 + dx^2.$$

For a future timelike segment $(\Delta t, \Delta x)$, the finite proper time is

$$\tau(\Delta t, \Delta x) = \sqrt{\Delta t^2 - \Delta x^2}.$$

Given endpoints $(0, 0)$ and (T, X) , the Pandrosion graph score is

$$\mathcal{J}_h(\Gamma) = \sum_{k=0}^{N-1} \sqrt{\Delta t_k^2 - \Delta x_k^2}, \quad |\Delta x_k| < \Delta t_k.$$

Proposition 2.1 (Flat-space calibration). *Among future timelike broken lines from $(0, 0)$ to (T, X) , with $|X| < T$, the proper-time score satisfies*

$$\sum_k \sqrt{\Delta t_k^2 - \Delta x_k^2} \leq \sqrt{T^2 - X^2}.$$

Equality holds exactly when all segments have the same velocity $\Delta x_k / \Delta t_k = X / T$. Thus the continuum Pandrosion score selects the inertial line.

Proof. The statement is the reverse triangle inequality for future timelike vectors. Equivalently, the function $v \mapsto \sqrt{1 - v^2}$ is concave on $(-1, 1)$, so Jensen's inequality gives the result after writing $\Delta x_k = v_k \Delta t_k$. $\square$

The implemented graph uses $T = 5$, $X = 1.2$, $N = 40$ time slices and spatial meshes from 0.1 down to 0.01250. The finest run gives

$$\mathcal{T}_{\text{exact}} = 4.853864, \quad \mathcal{T}_h = 4.847266, \quad |\mathcal{T}_{\text{exact}} - \mathcal{T}_h| = 6.598346e - 03.$$

A vanishing tie-breaker selects the balanced representative among lattice paths with equal leading proper-time score. This tie-breaker is diagnostic; the invariant leading score is the proper time above.

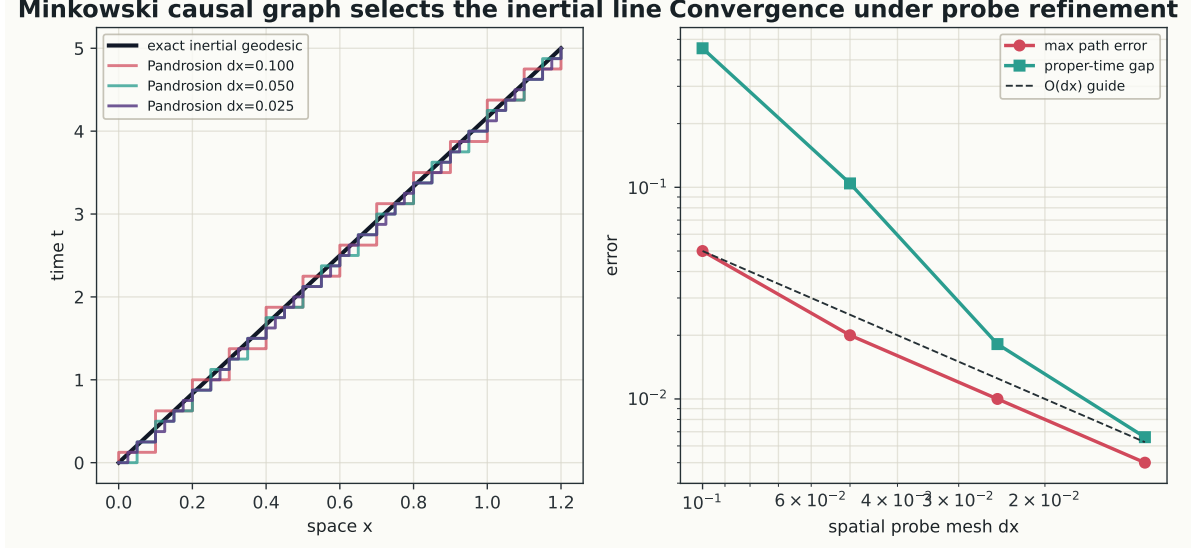


Figure 1: Minkowski benchmark. The causal graph maximizes finite proper time between fixed endpoints. The balanced lattice maximizer approaches the straight inertial worldline as the spatial probe mesh is refined.

3 Benchmark 2: Newtonian shadow

Restore c and use a weak static metric

$$ds^2 = - \left(1 + \frac{2\Phi(x)}{c^2} \right) c^2 dt^2 + dx^2 + \mathcal{O}(c^{-2} \|dx\|^2).$$

For slow motion,

$$d\tau = \left[1 + \frac{\Phi(x)}{c^2} - \frac{\dot{x}^2}{2c^2} \right] dt + \mathcal{O}(c^{-4}).$$

After dropping the constant factor, the Pandrosion weak-field score is

$$\mathcal{J}_h^{\text{Newt}}(\Gamma) = \sum_k \Delta t \left[\Phi \left(\frac{x_k + x_{k+1}}{2} \right) - \frac{1}{2} \left(\frac{x_{k+1} - x_k}{\Delta t} \right)^2 \right].$$

Proposition 3.1 (Newtonian bridge). *If $\Phi(x) = gx$, the continuum extremal of the weak-field Pandrosion score with fixed endpoints is*

$$x(t) = x_0 + v_0 t - \frac{1}{2} g t^2, \quad v_0 = \frac{x_1 - x_0 + \frac{1}{2} g T^2}{T}.$$

It satisfies Newton's equation $x'' = -g$.

Proof. The continuum score is

$$\int_0^T \left(\Phi(x) - \frac{1}{2} \dot{x}^2 \right) dt.$$

The Euler–Lagrange equation is

$$\frac{\partial \Phi}{\partial x} + \ddot{x} = 0.$$

For $\Phi(x) = gx$, this is $\ddot{x} = -g$, giving the stated bridge after enforcing the endpoints. $\square$

The numerical test uses $g = 0.350$, $T = 2.4$, $x_0 = 0.8$, and $x_1 = 0$. At the finest mesh $\Delta x = 0.00500$, the maximum position error is 0.01200 and the RMS error is 0.00851.

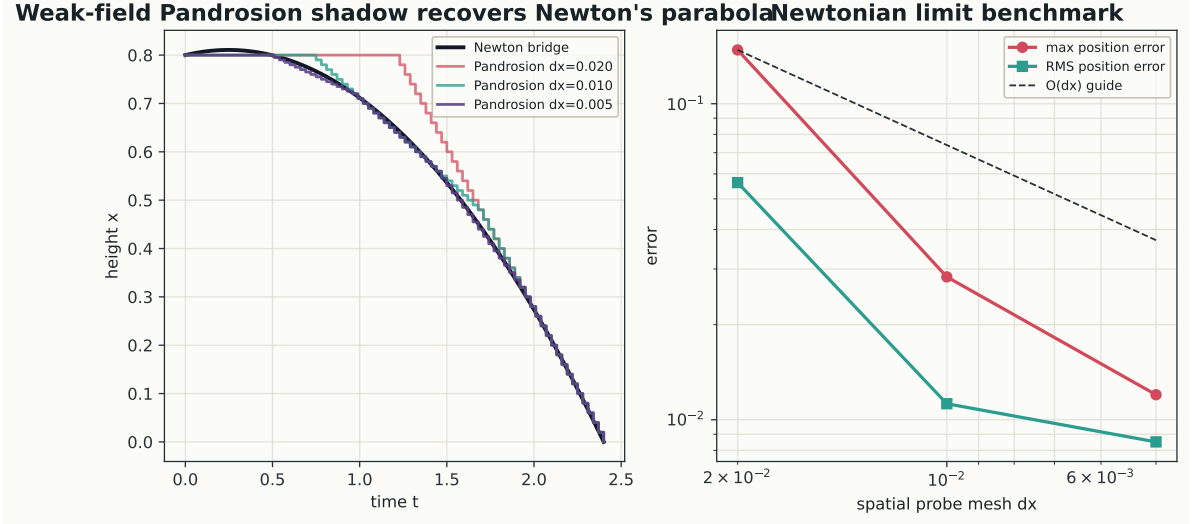


Figure 2: Newtonian shadow benchmark. The finite weak-field Pandrosion score selects a path converging to the constant-field Newtonian bridge.

4 Benchmark 3: horizon-safe dynamic atlas

The third test checks that a Pandrosion atlas distinguishes physics from bad coordinates. Consider radial Schwarzschild geometry with $M = 1$. In Schwarzschild coordinates,

$$f(r) = 1 - \frac{2M}{r}, \quad ds^2 = -f(r) dt^2 + f(r)^{-1} dr^2,$$

so the coordinate description is singular at $r = 2M$. In Painleve–Gullstrand time T ,

$$ds^2 = -f(r) dT^2 + 2\sqrt{\frac{2M}{r}} dT dr + dr^2,$$

which is regular at the horizon.

For radial free fall from rest at infinity,

$$\frac{dr}{dT} = -\sqrt{\frac{2M}{r}}, \quad r(T) = \left(r_0^{3/2} - \frac{3}{2}\sqrt{2M}T\right)^{2/3}.$$

In the Painleve–Gullstrand chart this path has $d\tau = dT$. The finite segment estimator therefore gives

$$\sum_k \tau_{h,PG} = 4.447376, \quad |T_{\text{end}} - \sum_k \tau_{h,PG}| = 9.059420e - 14.$$

Definition 4.1 (Chart condition score). *For the horizon benchmark use the model costs*

$$C_{\text{Sch}}(r) = 1 + \frac{0.02}{|f(r)|^2}, \quad C_{\text{PG}}(r) = 1.32 + 0.08 \left(\frac{2M}{r}\right)^2.$$

The dynamic Pandrosion atlas selects the chart with smaller condition cost.

In the run shown in Figure 3, the atlas switches at

$$r \simeq 2.603M, \quad T \simeq 3.290.$$

This is the numerical analog of the Riemann-chart logic in monomial Pandrosion: when one coordinate representation becomes unstable, the method changes chart while preserving the underlying invariant.

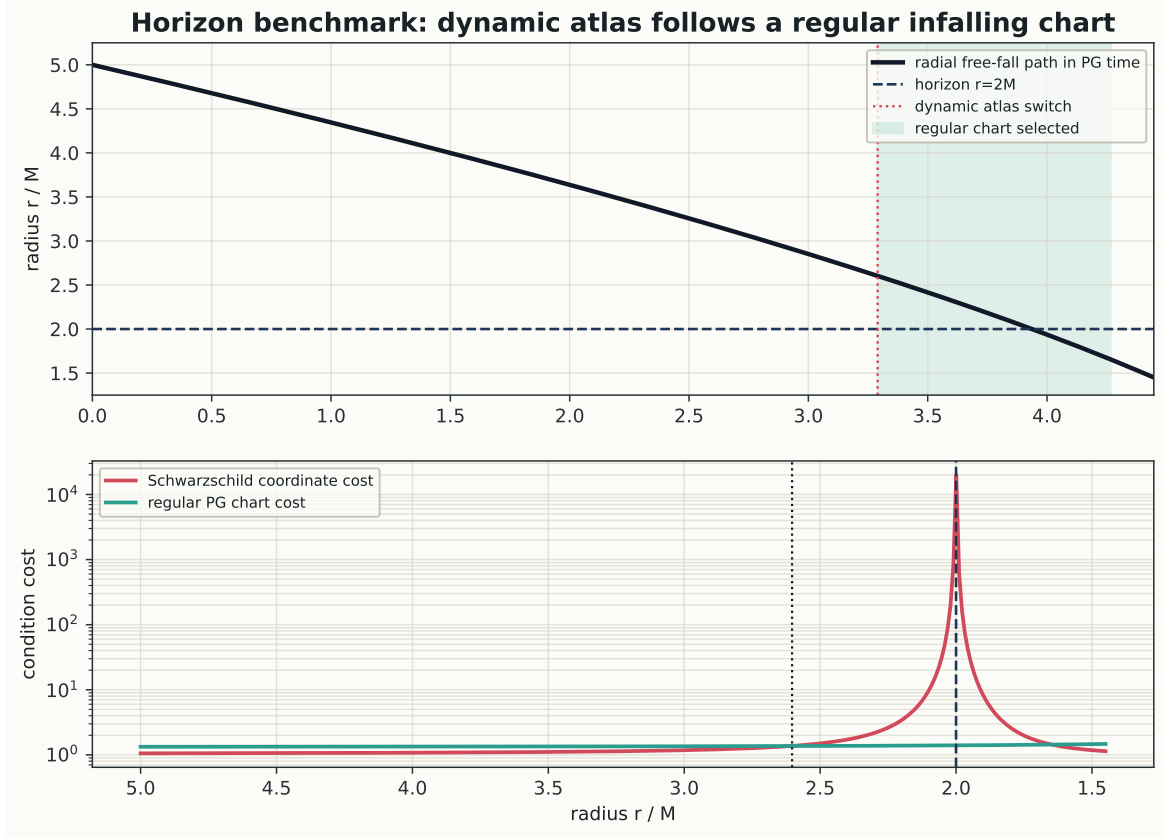


Figure 3: Horizon atlas benchmark. The physical infall path is regular, but the Schwarzschild coordinate cost diverges near $r = 2M$. The dynamic Pandrosion atlas switches to the regular Painleve–Gullstrand chart before the horizon.

Test	Classical invariant	Pandrosion result
Minkowski	inertial line and proper time	causal graph path error 0.00500, proper-time gap $6.598346e-03$ at $\Delta x = 0.01250$
Weak field	Newton equation $x'' = -g$	finite score recovers the parabolic bridge with RMS error 0.00851 at $\Delta x = 0.00500$
Horizon chart	regular infall across $r = 2M$	atlas switches to PG chart at $r \simeq 2.603M$ and keeps proper-time error $9.059420e-14$

Table 1: The three validation layers for Pandrosion relativity.

5 Benchmark summary

The results support the narrow claim of Paper 027. Pandrosion is not replacing relativity. It gives a finite-probe calculus whose leading score reproduces relativistic and Newtonian invariants when the graph and chart logic are consistent.

6 What remains to prove

The benchmarks suggest four precise next targets:

- (i) prove convergence rates for the causal graph score with finite tie-breakers and acceleration penalties;
- (ii) extend the weak-field test to nonconstant potentials and compare with standard symplectic variational integrators;
- (iii) test horizon charting with full geodesic optimization rather than a known analytic radial infall path;
- (iv) combine finite holonomy from Paper 027 with stress-energy sampling to test an Einstein-tensor balance on small lattices.

7 Conclusion

Paper 028 gives the first computational validation layer for Pandrosion relativity. The method passes three sanity checks: it recovers inertial motion in Minkowski space, recovers Newton’s equation in the weak-field slow-motion limit, and avoids a horizon coordinate singularity by dynamic atlas switching. The original contribution is not a new law of physics; it is the Pandrosion protocol that treats these classical laws through causal finite-slope probes and chart-invariant scoring.

References

- [1] R. M. Wald, *General Relativity*, University of Chicago Press, 1984.
- [2] B. O’Neill, *Semi-Riemannian Geometry*, Academic Press, 1983.
- [3] T. Regge, *General relativity without coordinates*, Nuovo Cimento 19, 558–571, 1961.
- [4] J. E. Marsden and M. West, *Discrete mechanics and variational integrators*, Acta Numerica 10, 357–514, 2001.
- [5] I. Besevic, *Discrete Geodesic Probe Convergence for Pandrosion*, manuscript, May 2026.
- [6] I. Besevic, *Pandrosion Relativity: Finite-slope causal probes, dynamic atlases, and geodesic physical motion*, manuscript, May 2026.